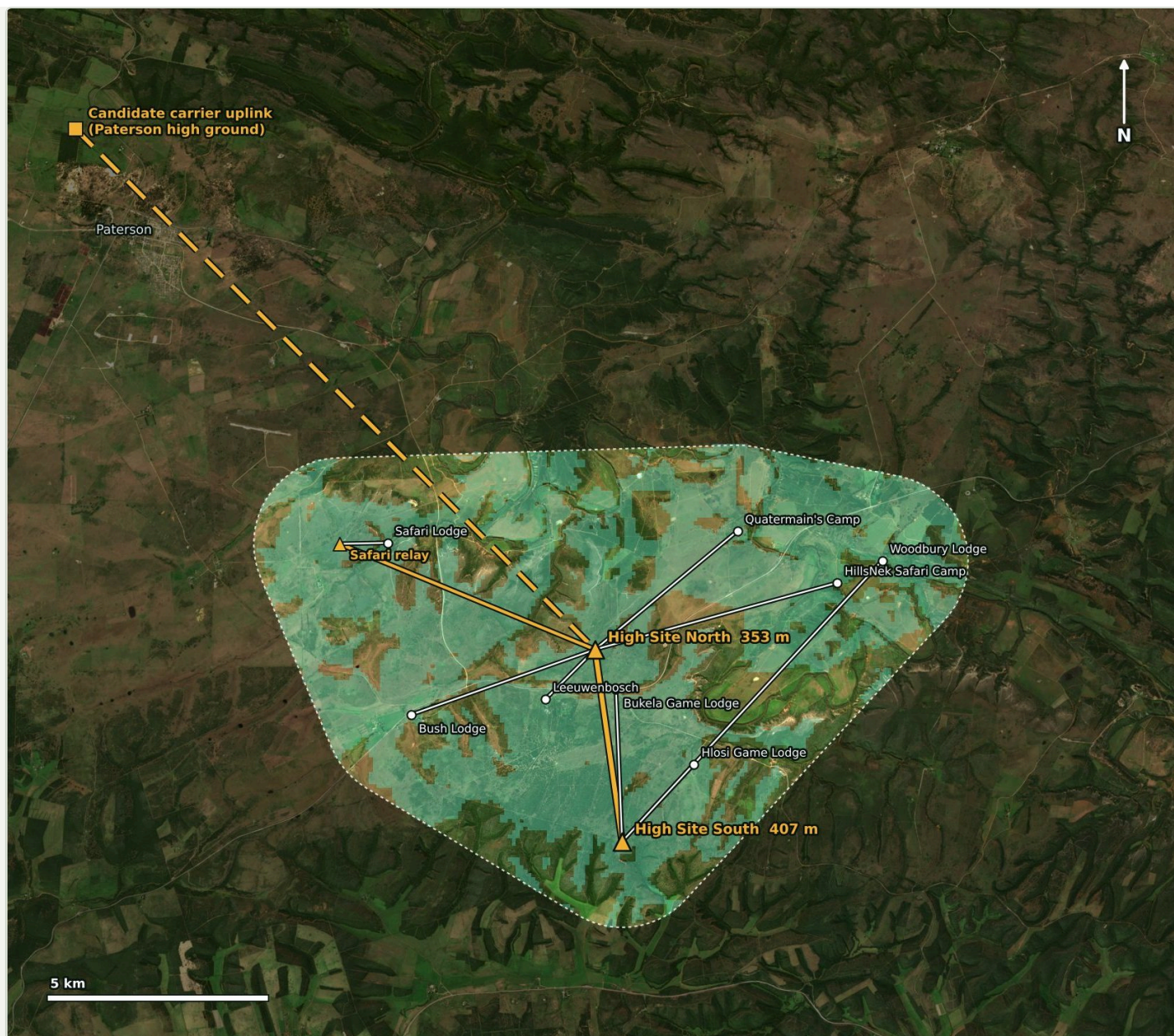


PREPARED FOR RICHARD, AMAKHALA GAME RESERVE

One network across the whole reserve.

We modelled Amakhala's terrain from satellite elevation data and ran every link through a line-of-sight and Fresnel-zone check. The result: **two ridge high-sites and one small relay put all eight lodges we could locate on clear radio paths**, fed from a carrier uplink on the high ground at Paterson.



--- Carrier uplink — Private backbone — Lodge link

■ Clear line of sight for a 3 m field terminal

Sentinel-2 imagery, 5 August 2026. The dotted outline is the study area around the lodges; everything outside it is dimmed. Site positions are desktop candidates, to be confirmed on a site visit.

8 of 8

lodges on a clear, Fresnel-zone-clear radio path

2 + 1

ridge high-sites plus one short relay for Safari Lodge in its valley

65%

of the 129 km² study area in line of sight for field teams and sensors

17.1 km

clear uplink path from Paterson high ground into the reserve

HOW IT IS BUILT

Carrier in, private network out

Most lodges on Amakhala list Wi-Fi in the public areas by the main lodge, with each lodge sorting out its own connection. We propose a single backbone that belongs to the reserve.

LAYER 1 • CARRIER

Uplink from Paterson

An uncontended business-grade Vodacom service, delivered by CTTX over a 17.1 km link from the high ground at Paterson to High Site North. LTE failover keeps the reserve connected if the carrier path drops.

LAYER 2 • BACKBONE

Two ridge high-sites

High Site North (353 m) and High Site South (407 m) sit 4.5 km apart with a clear link between them. Each runs on solar and lithium storage with remote monitoring, so the network stays up through load-shedding.

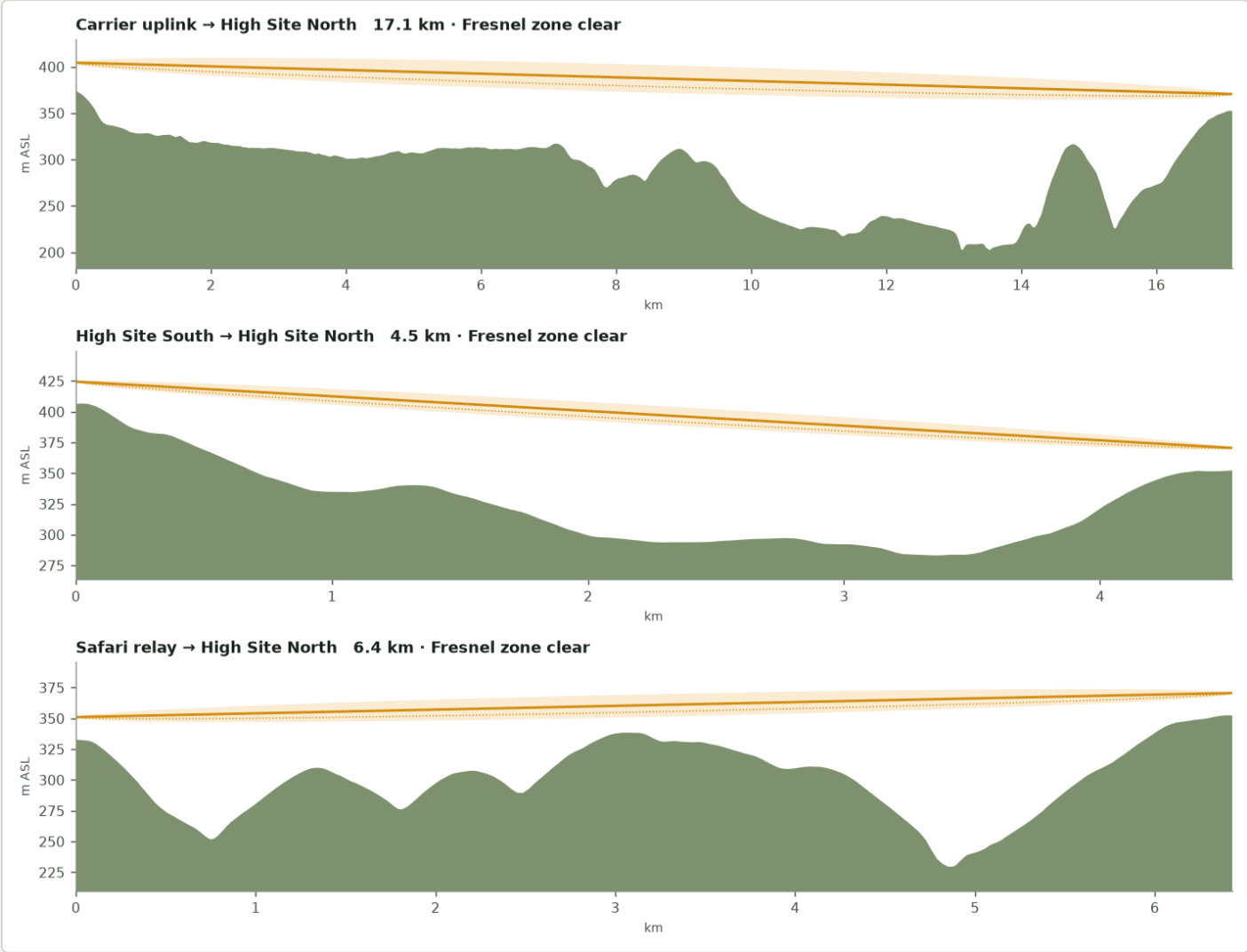
LAYER 3 • DISTRIBUTION

Every lodge and ops site

Sector radios on each high site reach the lodges, HQ, the anti-poaching base, ecology, the workshop and the gates. A LoRaWAN gateway on the backbone is ready for fence, water-point, pump and fuel sensors.

Every path checked against the terrain

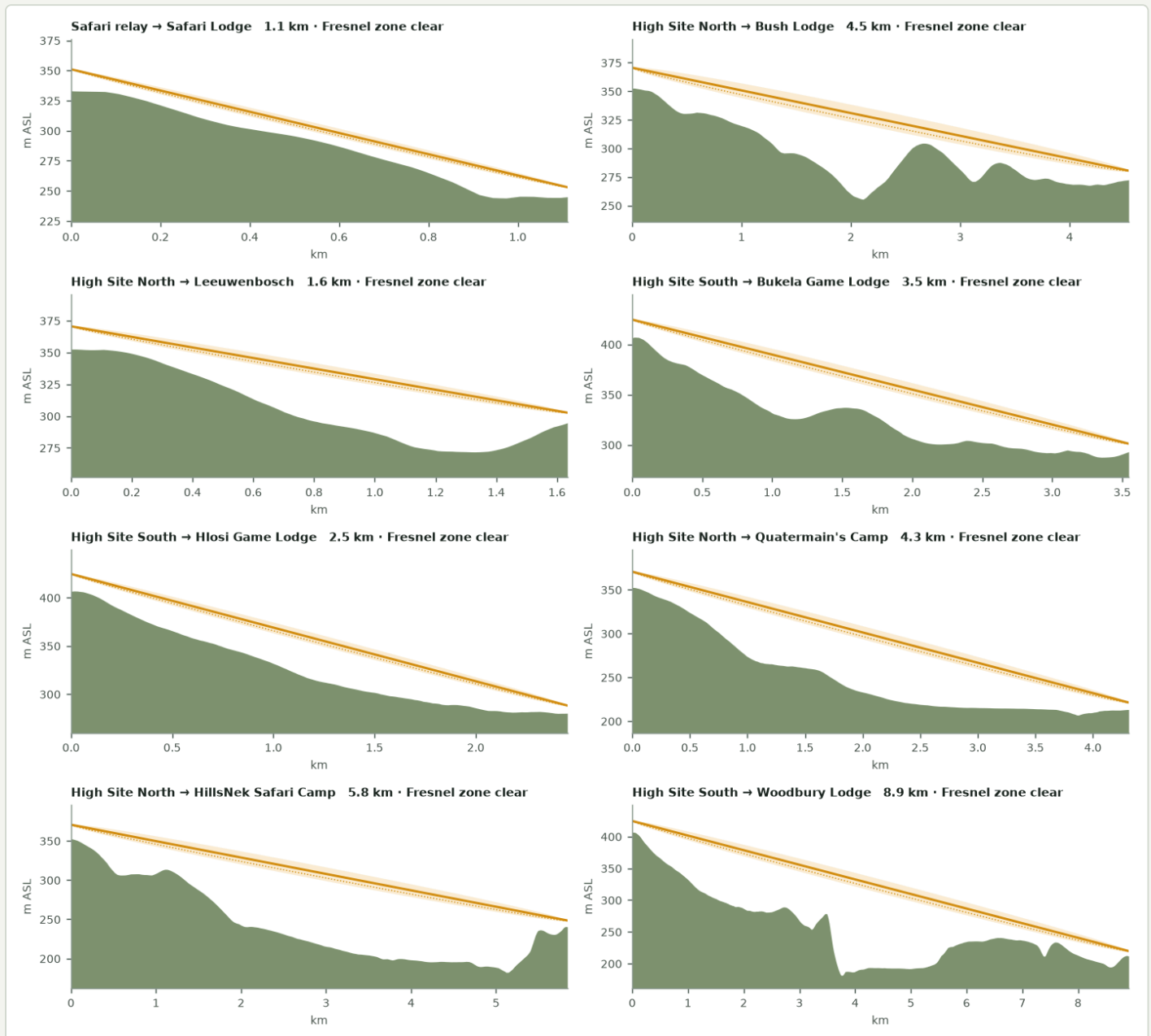
Each profile below shows the ground between two sites (green), the radio path (amber line) and its first Fresnel zone (amber band). We allow for earth curvature and require at least 60% of the Fresnel zone to be clear at every point. A path that fails is not drawn on the map.



Backbone and uplink paths. 5.8 GHz, $k = 4/3$, 30 m uplink tower, 18 m high-site masts.

LODGE	SERVED FROM	PATH LENGTH	LINE OF SIGHT
Safari Lodge	Safari relay	1.1 km	Clear
Bush Lodge	High Site North	4.5 km	Clear
Leeuwenbosch	High Site North	1.6 km	Clear
Bukela Game Lodge	High Site South	3.5 km	Clear

LODGE	SERVED FROM	PATH LENGTH	LINE OF SIGHT
Hlosi Game Lodge	High Site South	2.5 km	Clear
Quatermain's Camp	High Site North	4.3 km	Clear
HillsNek Safari Camp	High Site North	5.8 km	Clear
Woodbury Lodge	High Site South	8.9 km	Clear



Lodge paths, with an 8 m receiver mast at each lodge. Lodge positions come from published GPS coordinates.

What changes when the reserve owns its network

Today: a connection per lodge

- Each lodge buys and manages its own service
- Coverage stops at the main building
- Security, ecology and maintenance teams are not on a shared network
- Nothing links the gates, water points or fences

With a reserve backbone

- One uncontended carrier feed shared across the reserve
- Lodges, HQ, APU base and gates on one private network
- Coverage out into the field, day and night
- An owned asset that takes cameras and sensors without new towers

Avoided loss

Gate cameras, fence and trigger alarms, and after-hours communication for the Equine APU and K9 unit. When something happens at night, the right people know in minutes.

Operational efficiency

Water points, pumps and fuel tanks report their own status, so there are fewer vehicle trips just to check. Maintenance, ecology and lodges share one communications path.

Infrastructure ownership

High sites, backbone and power belong to the reserve. The network grows with Amakhala instead of depending on each lodge's separate provider.

Executive risk reduction

The reserve manager gets live visibility of the network, cameras and sensors, plus a record of operational diligence for owners, insurers and conservation partners.

NEXT STEP

Check it against what you see on the ground

30-minute call

Your peak usage, the incidents hardest to coordinate, and what currently needs a vehicle trip to check.

Confirm the sites

A marked map of HQ, the APU base and the gates. We add them to the model and rerun the link plan.

One day on the reserve

We walk the two ridges and the Safari relay point and confirm the carrier uplink with Vodacom.

Design and proposal

A final network design in Cambium LINKPlanner, with equipment, power and a phased plan.

Method. Terrain: Copernicus DEM GLO-30 (30 m). Imagery: ESA Sentinel-2 L2A true colour, 5 August 2026. Radio model: 5.8 GHz, effective earth radius factor 4/3, 60% first-Fresnel-zone clearance required at every point. Heights: carrier uplink 30 m, high-site masts 18 m, lodge masts 8 m, field terminals 3 m. The DEM does not model trees or buildings. Final paths are confirmed on survey and in Cambium LINKPlanner before anything is built.

Candidate positions. High sites North -33.52630, 26.08097, South -33.56667, 26.08733, Safari relay -33.50363, 26.01726, carrier uplink -33.41597, 25.95211. The uplink marks the high ground at Paterson. The exact carrier site is subject to Vodacom feasibility.